

Digital Image Analysis Tutorial 6

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1 """AIS31C T06 studio – Registration Lab (Tue 8 Sep, 2-3 PM).
2 INPUT: camera as the reference; a synthetically rotated(+7 deg)+shifted(12,-8) copy
3 | as the 'second date' – so ground truth is KNOWN.
4 OPERATION: recover the affine from 3 clean point pairs -> measure residual ->
5 | inject ONE corrupted pair (30 px off) into a 6-pair least-squares fit (watch it
6 | bend) -> rescue with RANSAC via estimateAffinePartial2D.
7 PARAMETERS: angle 7 deg; shift (12, -8); outlier offset 30 px; RANSAC thr 3.
8 EXPECTED OUTPUT: printed recoveries (asserted) + expected_outputs/T06_panel.png.
9 INTERPRETATION: least squares follows outliers (squared error!); RANSAC votes them out.
10 """
11 import numpy as np, cv2, os, matplotlib
12 matplotlib.use("Agg"); import matplotlib.pyplot as plt
13 from skimage import data
14 HERE = os.path.dirname(os.path.abspath(__file__))
15 os.makedirs(f"{HERE}/expected_outputs", exist_ok=True); os.makedirs(f"{HERE}/inputs", exist_ok=True)
16 img = data.camera()
17 Mtrue = cv2.getRotationMatrix2D((256, 256), 7, 1.0); Mtrue[:, 2] += (12, -8)
18 moved = cv2.warpAffine(img, Mtrue, (512, 512))
19 cv2.imwrite(f"{HERE}/inputs/reference.png", img); cv2.imwrite(f"{HERE}/inputs/moved.png", moved)
20
21 pts = np.array([[100,100],[400,120],[250,400],[120,380],[420,380],[256,200]], np.float64)
22 dst = (Mtrue[:, :2] @ pts.T).T + Mtrue[:, 2]
23
24 # [1] exact recovery from 3 pairs
25 M3 = cv2.getAffineTransform(pts[:3].astype(np.float32), dst[:3].astype(np.float32))
26 ang = np.degrees(np.arctan2(M3[1,0], M3[0,0]))
27 print(f"[1] 3-pair recovery: angle {ang:.2f} deg (truth 7.00), tx {M3[0,2]:.1f} ty {M3[1,2]:.1f}")
28 assert abs(ang - (-7)) < 0.05 or abs(ang - 7) < 0.05
29
30 # [2] one bad pair poisons PLAIN least squares (fit it ourselves, transparently)
31 bad = dst.copy(); bad[5] += (30, -30)
32 A = np.zeros((12, 6)); A[0::2, 0:2] = pts; A[0::2, 2] = 1; A[1::2, 3:5] = pts; A[1::2, 5] = 1
33 M6 = np.linalg.lstsq(A, bad.ravel(), rcond=None)[0].reshape(2, 3)
34 res6 = np.abs((M6[:, :2] @ pts.T).T + M6[:, 2] - dst).max()
35 print(f"[2] 6 pairs, one 30-px outlier, plain least squares: worst residual vs truth {res6:.1f} px")
36 assert res6 > 5 # the outlier bent the whole fit
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37
38 # [3] RANSAC rescue
39 Mr, mask = cv2.estimateAffine2D(pts.astype(np.float32), bad.astype(np.float32), method=cv2.RANSAC, ransacReprojThreshold=3.0)
40 resR = np.abs((Mr[:, :2] @ pts.T).T + Mr[:, 2] - dst).max()
41 print(f"[3] RANSAC (thr 3): kept {int(mask.sum())}/6 pairs, worst residual {resR:.2f} px")
42 assert int(mask.sum()) == 5 and resR < 1.0
43
44 back = cv2.warpAffine(moved, cv2.invertAffineTransform(Mr), (512, 512))
45 fig, ax = plt.subplots(1, 4, figsize=(13, 3.6))
46 for a, im, t in zip(ax, [img, moved, np.abs(img.astype(int)-moved.astype(int)), np.abs(img.astype(int)-back.astype(int))],
47     ["reference", "moved (+7 deg, shift)", "|difference| BEFORE", "|difference| AFTER RANSAC"]):
48     a.imshow(im, cmap="gray"); a.set_title(t, fontsize=13); a.axis("off")
49 plt.tight_layout(); plt.savefig(f"{HERE}/expected_outputs/T06_panel.png", dpi=120); plt.close()
50 print("saved expected_outputs/T06_panel.png")
51
52 def support():
53     print("[SUPPORT] use only step [1]: three clean pairs, cv2.getAffineTransform.")
54     print(" Verify the recovered angle is ~7 deg. Then change ONE point by 5 px and watch the angle move.")
55
56 def extension():
57     h = np.eye(3); h[2, 0] = 2e-4 # mild perspective
58     pers = cv2.warpPerspective(img, h @ np.vstack([Mtrue, [0,0,1]]), (512, 512))
59     cv2.imwrite(f"{HERE}/inputs/perspective.png", pers)
60     print("[EXTENSION] inputs/perspective.png has mild PERSPECTIVE. Show the best affine fit")
61     print(" cannot flatten the residual, then recover it with cv2.findHomography (4+ pairs).")
62
63 if __name__ == "__main__":
64     import sys
65     if len(sys.argv) > 1 and sys.argv[1] == "support": support()
66     elif len(sys.argv) > 1 and sys.argv[1] == "extension": extension()

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Output Images:

reference



moved (+7 deg, shift)



|difference| BEFORE



|difference| AFTER RANSAC

